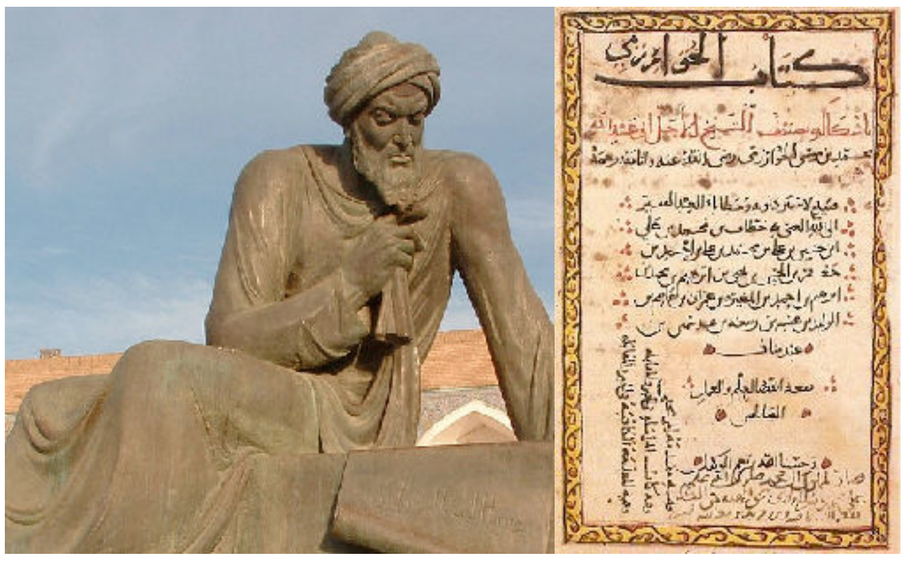
25. What Is an Algorithm?

The Algorithm Concept



Muhammad ibn Mūsā al-Khwārizmī (708-850), Baghdad



Ada Lovelace (1815-1852), London



Alan Turing (1912-1954), Wilmslow, England

An **algorithm** is a *procedure* such that:

1. The procedure can be divided into finitely many steps.
2. The transition from one step to another is completely determinate.

A **program** is a *description* of an algorithm in a given computational language. But a machine can **implement** an algorithm in its physical circuitry, without a program. This is the case of finite state machines that are based on sequential circuits.

Steps in the algorithm are the same as states in an FSM. States and state transitions are the basic units out of which algorithms are built.

To do *X*

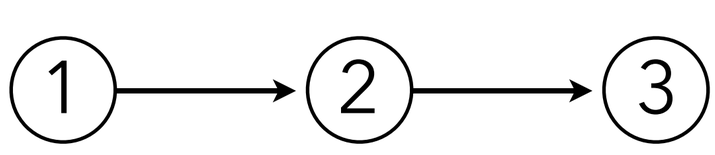
1. [redacted]
2. [redacted]
3. [redacted]

To do *X*

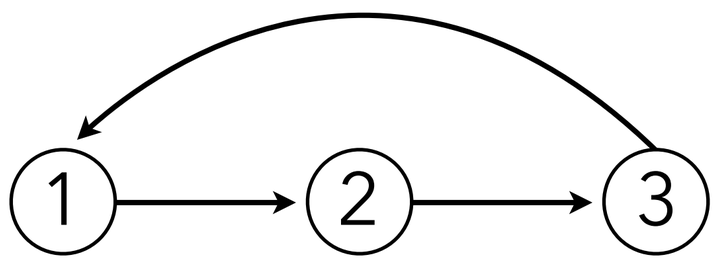
1. [redacted]
2. [redacted]
3. [redacted]

and repeat State 1.

Running an algorithm on a given sequence of inputs involves a certain number of transitions. This is the total number of state transitions that occur until the input is fully processed.



In this case, there are only three states, and only three possible steps.



In this case, there are only three states, but potentially infinite number of transitions.

Sorting

Branching search algorithm: guess a number between 1 and 100.

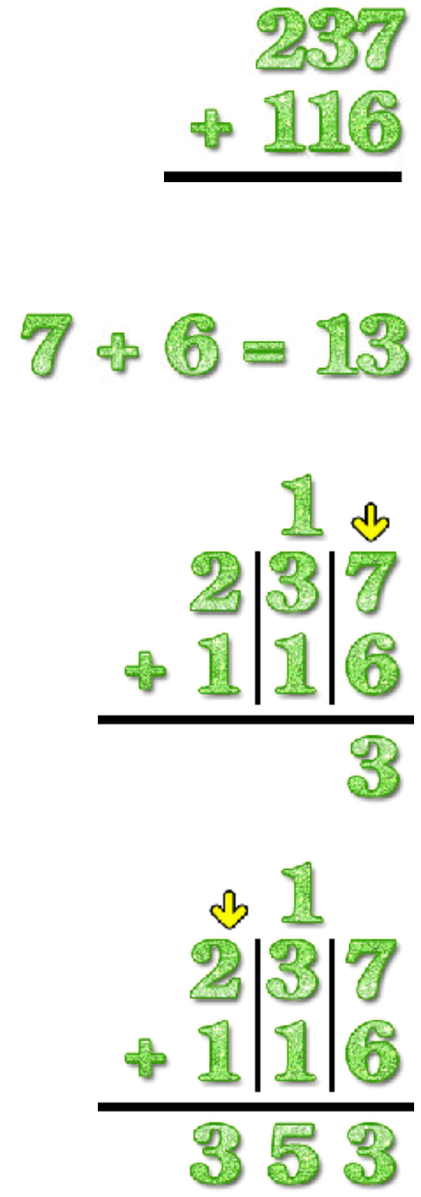
1. Set **Low**=1 and **High**=100.
2. Set **Guess** to the middle number between **Low** and **High** (round down).
3. Collect answer.
   a. If the answer is "Correct," stop.
   b. If the answer is "Higher," set **Low** = **Guess**+1.
   c. If the answer is "Lower," set **High** = **Guess**-1.
4. Go back to Step 2

Note! Search by counting up from 1 could take up to 100 steps. The branching search algorithm always takes under 7 steps.

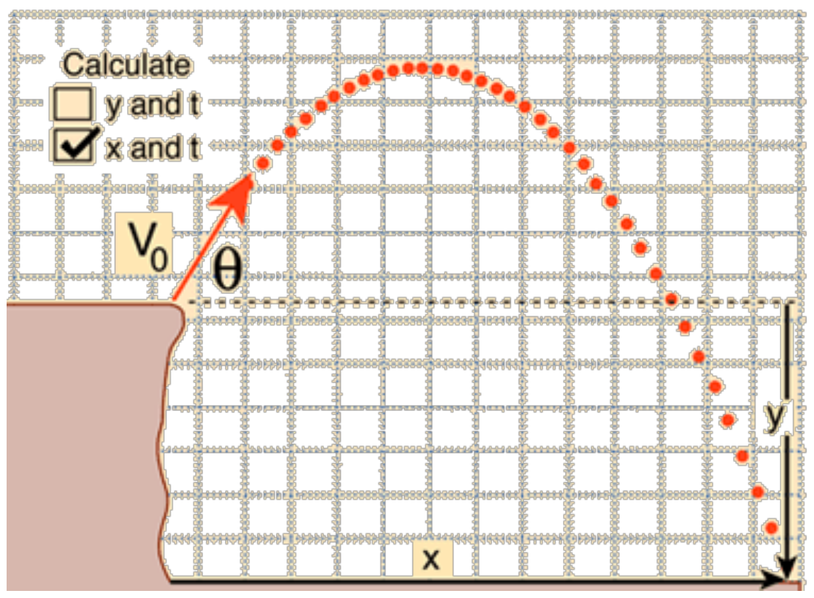
Math

How to Do Long Addition

1. Write down the numbers you wish to add, one underneath the other.
2. Add up the numbers in the right-most column.
3. Check if the answer from the previous step is 9 or less:
   1. If Yes, write the number below the column and move to Step 4.
   2. If No, the answer will have two digits. Write the digit on the right below the column. Write the digit on the left above the column to the left of the current column. Move to Step 4.
4. Move to the column to the left. Add up the numbers in this column, including any numbers above the two lines.
5. Go to Step 3 and repeat until all columns have been added.



Physics



$$x = v_{0x}\,t$$

$$y = v_{0y}\,t - \frac{1}{2}gt^2$$

Using the **quadratic formula** to solve for t gives two values of time for a given value of y:

$$t = \frac{v_{0y}}{g} \pm \sqrt{\frac{v_{0y}^2}{g^2} - \frac{2y}{g}}$$

Substitution of the two time values gives the two values of x corresponding to a given height y.

Calculation

Algorithms and Local Logics

Following an algorithm succeeds as an evolutionary strategy to the extant that it can mirror a relevant process in the world.

Perhaps an ideal being could simulate everything in the world by implementing algorithms that perfectly implemented the laws of physics. The laws of physics are a **global logic**, a set of rules that describes all situations.

We succeed to the extent that we can identify constrained situations governed by **local logics**. Examples of local logics are the laws of psychology, the laws of driving, the laws of boiling an egg.

An algorithm for boiling eggs. The algorithm is possible because the behavior of eggs is highly predictable— it follows a local logic.

1 Turn on the stove
2 Fill the pan with water
3 Place the pan on the stove
4 When the water boils, add one egg, and set the timer
5 When the timer rings, turn off the gas
6 Remove the egg from the water
7 Result: one boiled egg.

